Structural Instability of the Phenomenological Null Regime in Causally Rigid Channels

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Abstract

We introduce an abstract phenomenological space $\mathcal{E}$ and a compatibility operator $\Phi : Z \to \mathcal{E}$ on a declared metric comparison domain. We define a null regime $\varnothing_\phi \in \mathcal{E}$ and prove a **Phenomenological Instability Theorem**: under a certified positive lower readout bound, two defined, separated witness values cannot both equal the null element. CCR is a possible application class and is not needed for this metric implication. Consequently, a CCR realization with such a witness induces a witness-relative non-null separation. The result is *negative-structural*: it does not posit experience; it forbids one typed joint-null configuration under stated hypotheses. We provide explicit coupling bounds in two canonical families (profinite and symbolic dynamics) and prove factorization on a faithful image and a finite-prefix approximation bound. The results are metric, order-theoretic and representation-relative; physical interpretation requires additional evidence.

1 Scope, System Boundary, and Non-Inference Note

Scope and admissible reading. This paper proves a witness-relative negative-structural result: under CCR constraints plus a separately typed extension witness, a lower Lipschitz constant, and an admissible compatibility operator, joint exact collapse to the null regime is excluded. It develops explicit bounds, factorization structure, and canonical-family consequences for that conditional result. It does not define or detect consciousness, identify $\mathcal{E}$ with subjective experience, assert that CCR systems exist in nature, or derive ethical or metaphysical closure from the theorem. Related systems may implement admissibility-governed diagnostics, boundary-sensitive stress tests, and runtime comparisons motivated by this paper, but those outputs do not implement the abstract phenomenological space $\mathcal{E}$ and do not convert current runtime behavior into proof that any concrete system instantiates experience.

Witness-relative clarification. CCR by itself does not provide the extension witness, the lower Lipschitz constant, or the admissible compatibility operator. The historical phrase "null instability" denotes joint exact null exclusion here, not temporal instability. Statements of non-nullity use the explicit hypotheses of Theorem 5.2, including the declared witness domain of Definition 5.1. The null-instability theorem must not be read as a consciousness detector, an AI-subjectivity claim, a moral-status claim, an empirical certificate, or validation of the corpus by the runtime.

2 Introduction

2.1 Motivation and Scope

Papers I and II established a mathematical framework for structural identity as an inverse limit and classified phenomenological regimes compatible with rigid identity. This paper addresses the fundamental question:

Can a CCR system remain in a structurally null phenomenological state?

We exclude joint null assignment on a separated, detected witness pair. This does not exclude a null value at one endpoint or a dynamically stable null state.

2.2 Main Result (Informal)

No-Null Regime Theorem: For a CCR channel equipped with the separated extension witness and regularity hypotheses stated in Theorem 5.2, joint exact assignment to the null element $\varnothing_\phi$ is excluded. At least one endpoint is non-null; positive distance from null and a gap to every lower regime are distinct conclusions.

Example 2.1 (Nonempty Assignment and Null-Anchored Witness)**.** Let $Z = \{0,1\}^{\mathbb{N}}$, let $d_Z(x,y) = 3^{-(m+1)}$ at their first differing bit m, and let $\mathcal{E} = [0,1]$ with its usual order, metric and null 0. Set $d_Z(x,x) = 0$. The ternary readout $F(b) = \sum_{j\geq 0} 2b_j/3^{j+1}$ satisfies

$$d_Z(x,y) \leq |F(x) - F(y)| \leq 3d_Z(x,y).$$

The first differing term has magnitude $2/3^{m+1}$ and the remaining tail is at most $1/3^{m+1}$, proving both bounds. The all-zero and all-one sequences give images 0 and 1, a null-anchored pair. Every finite prefix has different readout completions, while uniform tail error is $3^{-(N+1)}$; thus the revised E1 and E2 are realized together. The finite positive witness image $\{1\}$ has relative gap one, but the ambient global lower gap at 1 is zero. This model supplies readout geometry, not an infinite-mass cost certificate or phenomenal evidence.

2.3 Paper Organization

Section 2 reviews prerequisites from Papers I–II. Section 3 defines the phenomenological space with full axiomatic clarity. Section 4 proves the main instability theorem. Section 5 stratifies positive regimes. Section 6 provides quantitative bounds. Section 7 combines the conditional witness conclusions. Section 8 explicitly delimits scope. Section 9 compares with existing frameworks. Section 10 discusses limitations. Section 11 concludes. Appendices provide canonical examples with explicit computations.

2.4 What This Paper Adds Beyond Phenomenological Regimes

Phenomenological Regimes classifies the admissible regime space and derives the continuity and non-constancy constraints that any compatible assignment must satisfy. This paper takes the remaining case—the null regime itself—and proves joint null exclusion under independently supplied witness hypotheses.

3 Preliminaries from Papers I–II

3.1 Identity as Inverse Limit

From *Rigid Identity*, we have:

Definition 3.1 (Identity Object)**.** Let $(S_t, \pi_{t+1\to t})_{t\in T}$ be a projective system of state spaces with surjective, compatible projections. The **identity object** is:

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_t) \in \prod_t S_t : \pi_{t+1\to t}(x_{t+1}) = x_t \right\}$$

Definition 3.2 (Weighted Deformation Metric)**.** For $x, y \in \mathcal{I}$, use $d_w(x,y) = \sup_t w_t d_t(x_t, y_t)$ with $w_t > 0$, $\sum_t w_t < \infty$, and normalized complete coordinate metrics $0 \leq d_t \leq 1$. The normalization guarantees a finite metric and uniformly vanishing tails. This metric is on histories, not on the set of identity objects of different systems.

Proposition 3.3 (Metric Completeness, Rigid Identity)**.** *With the normalized complete coordinate metrics above and continuous projective maps, $(\mathcal{I}, d_w)$ is complete, when non-empty.*

Proof. A d_w-Cauchy sequence is Cauchy in each coordinate, since $w_t > 0$. Coordinate completeness supplies limits; continuity of each projection preserves compatibility. For convergence in d_w, first choose a finite prefix with $\sup_{t>N} w_t$ small, then control the finitely many remaining coordinates. The normalized tail is uniformly small. □

3.2 Ontological Mass and CCR Classification

Definition 3.4 (Ontological Mass, infimum convention)**.** Specify a metric comparison space (Z, d_Z) containing a chosen point z_S for each object under comparison and a declared intervention sector with cost $E_w(\delta) \in [0, \infty]$. Define

$$M_\Omega(S) = \inf \left\{ \frac{E_w(\delta)}{d_Z(z_S, z_{S,\delta})} : \delta \text{ admitted}, \ 0 < d_Z(z_S, z_{S,\delta}) < \infty \right\},$$

with $\inf \varnothing = +\infty$. A common metric/embedding and cost model must be supplied; $\mathcal{I}$ itself is a set, not an input to a map whose domain is that set.

Remark 3.5 (v24 consistency patch). Earlier drafts used a supremum notation for M_Ω in this paper. That notation is superseded here. The infimum convention measures a lower resistance to deformation in a specified sector, not a largest observed cost. All claims below are read under this convention.

Definition 3.6 (CCR Channel)**.** A channel is **Closed Causally Rigid (CCR)** if $M_\Omega = +\infty$.

Remark 3.7 (Physical Interpretation). $M_\Omega = +\infty$ refers to the declared intervention/cost model: there is no finite-cost nonzero deformation in that sector. Geometry alone supplies neither an energy unit nor a physical realization.

3.3 Key Theorems from Papers I–II

Theorem 3.8 (Finite Representation Budget)**.** *Fix a finite test set $A \subseteq Z$ with $d_{\mathcal{E}}(\Phi(x), \Phi(y)) \geq \delta > 0$ for every distinct $x, y \in A$. An encoder $q : A \to Q$ with $|Q| < |A|$ and any decoder $g : Q \to \mathcal{E}$ incur worst-case error at least $\delta/2$. Finite real dimension alone is not this cardinal budget.*

Proof. Two distinct points share an encoder value by the pigeonhole principle. If both reconstruction errors were smaller than $\delta/2$, the triangle inequality would make their true readout distance smaller than δ. □

Theorem 3.9 (Audited Linear Coupling)**.** *If an audited linear readout T from a nonzero finite-dimensional Euclidean space to another Euclidean space is injective, then its least singular value is positive and $\|T(x-y)\| \geq \sigma_{\min}(T)\|x-y\|$. A nonlinear readout requires a separate lower-bound proof; a named spectral gap does not supply one.*

Proof. On the compact unit sphere, the continuous function $v \mapsto \|Tv\|$ attains a strictly positive minimum when T is injective. Scaling gives the bound. This is a readout singular-value statement, not an inference from the dynamics spectrum. □

Theorem 3.10 (Continuity under Controlled Approximation)**.** *In a complete metric codomain, suppose finite continuous readouts f_n satisfy $d(f_{n+1}(z), f_n(z)) \leq a_n$ uniformly, with $\sum_n a_n < \infty$. Their limit Φ is continuous. The mass label is not sufficient by itself.*

Proof. The tail bound $d(\Phi(z), f_N(z)) \leq \sum_{n \geq N} a_n$ is uniform. Given an error, choose this tail small, use continuity of f_N, and apply the triangle inequality at the two endpoints. □

Remark 3.11 (Remaining null-regime gap). *Phenomenological Regimes* classifies admissible phenomenological regimes and derives the continuity constraints that govern them, but a null label alone does not supply an instability result. The task here is to derive the precise witness-relative conclusion and its limits.

The results below address joint null exclusion; temporal attraction and existence of physical CCR systems remain separate obligations.

4 The Phenomenological Space

4.1 Abstract Definition

Definition 4.1 (Phenomenological Space)**.** A **phenomenological space** is a quadruple $(\mathcal{E}, \preceq, d_{\mathcal{E}}, \varnothing_\phi)$ where:

1. $(\mathcal{E}, \preceq)$ is a partially ordered set (poset).
2. $(\mathcal{E}, d_{\mathcal{E}})$ is a complete metric space.
3. $\varnothing_\phi := \inf \mathcal{E}$ exists and is unique (the **null regime**).
4. **Order-metric compatibility:** if $e_1 \preceq e_2$, then $d_{\mathcal{E}}(e_1, \varnothing_\phi) \leq d_{\mathcal{E}}(e_2, \varnothing_\phi)$.
5. **Separation:** for $e \neq \varnothing_\phi$, we have $d_{\mathcal{E}}(e, \varnothing_\phi) > 0$.

Remark 4.2 (Interpretive Neutrality). $\mathcal{E}$ is an abstract regime space. No phenomenological content is assumed. The null regime $\varnothing_\phi$ represents structural absence, not "lack of experience." The term "phenomenological" is inherited from *Phenomenological Regimes* and refers to the coupling structure, not to philosophy of mind.

Remark 4.3 (Null regime versus undefined assignment). The symbol $\varnothing_\phi$ serves two distinct roles in the broader framework, and conflating them creates a category error:

1. **Null regime ($\varnothing_\phi \in \mathcal{E}$):** In Definition 4.1, $\varnothing_\phi$ is the bottom element of the phenomenological poset $(\mathcal{E}, \preceq)$. It is a legitimate element of $\mathcal{E}$, comparable to all other regimes via $\preceq$ and separated from them by $d_{\mathcal{E}}$.
2. **Undefined assignment ($\perp \notin \mathcal{E}$):** When a compatibility operator Φ is not defined on some input $x \in Z$, we write $\Phi(x) = \perp$. The symbol $\perp$ is disjoint from $\mathcal{E}$; it lives in the typed codomain $\mathcal{E}_\perp := \mathcal{E} \sqcup \{\perp\}$.

The instability theorem (Theorem 5.2) concerns admissible values in $\mathcal{E}$ and rules out two separated witness endpoints both taking the value $\varnothing_\phi$. The v31 conditional closure theorem concerns a different case: Φ undefined on both witness endpoints, written $\Phi(z_0) = \perp$ and $\Phi(z_1) = \perp$. That case is excluded by the witness hypothesis, which requires Φ to be defined on both endpoints.

Non-Theorem 4.4 (What $\mathcal{E}$ Is Not). $\mathcal{E}$ is not:

- A space of qualia or subjective states
- A model of consciousness
- A physical or neural state space
- An interpretation of quantum mechanics

$\mathcal{E}$ is a mathematical object satisfying Definition 4.1. All interpretive mappings are external to the framework.

4.2 Compatibility Operator

Definition 4.5 (Compatibility Operator)**.** For each system declare a metric domain D_S and a map $\Phi_S : D_S \to \mathcal{E}$. Audited comparisons use a single complete metric domain (Z, d_Z) and its specified readout $\Phi : Z \to \mathcal{E}$. One may take $Z = \mathcal{I}$ for histories; comparison of whole identity objects instead requires an explicit metric representation of those objects.

(C1) For a declared extension transport $j : D_S \to D_{S'}$, extension invariance means $\Phi_{S'} \circ j = \Phi_S$, in the common codomain.

(C2) For a declared isomorphism transport $u : D_S \to D_{S'}$, isomorphism stability means $\Phi_{S'} \circ u = \Phi_S$.

(C3) On the declared audited pair class $W \subseteq Z \times Z$, a certified $C > 0$ satisfies $d_{\mathcal{E}}(\Phi(x), \Phi(y)) \geq C d_Z(x, y)$.

Exact transport of a point under C1/C2 is not a new separated witness. In an extension domain a second, distinct point must be supplied. The pair theorem uses C3 only; it does not derive the maps or the witness from CCR.

Remark 4.6 (Partial definedness and the $\perp$ marker). In contexts where Φ is only partially defined, $\Phi(x) = \perp$ means that no admissible phenomenological assignment in $\mathcal{E}$ is available at x. The hypotheses (C1)–(C3) above apply only to inputs whose Φ-values lie in $\mathcal{E}$. The lower Lipschitz bound (C3) in particular requires both $\Phi(x), \Phi(\tilde{x}) \in \mathcal{E}$.

Lemma 4.7 (Well-Posedness of (C3))**.** *Continuity and a positive lower bound are jointly realizable; for example $Z = \mathcal{E} = [0, 1]$ with the ordinary metric and $\Phi(x) = x$ satisfies both with $C = 1$.*

Proof. The identity map preserves distances exactly. Neither condition is deduced from CCR. □

Remark 4.8. Axioms (C1)–(C3) are structural. No phenomenological content is introduced.

4.3 Attractor Set

Definition 4.9 (Attractor Set)**.** Let the allowed readout perturbations and extension transports be specified as below. Define $\mathrm{Attr}(S)$ as their structurally stable readout values. This notation does not assert temporal attraction; a flow and a basin would be additional data.

Definition 4.10 (Stability)**.** A value $e = \Phi_S(z)$ is structurally stable if some finite modulus $\omega(\rho) \to 0$ as $\rho \downarrow 0$ satisfies $d_{\mathcal{E}}(\Phi_{S'}(Pz), e) \leq \omega(\rho)$ for every declared perturbation transport $P : D_S \to D_{S'}$ of cost at most ρ. Exact persistence would instead require equality. For $\Phi(x) = x$ on $[0, 1]$ with clipped additive perturbations, the null value 0 is stable with $\omega(\rho) = \rho$, although a separated point has a non-null image.

4.4 The Bridge Lemma Revisited

From *Phenomenological Regimes*, we have:

Lemma 4.11 (Bridge Lemma)**.** *For any defined $e = \Phi(z) \in \mathcal{E}$, $d_{\mathcal{E}}(e, \varnothing_\phi) = 0$ implies $e = \varnothing_\phi$. No CCR assumption is required.*

Proof. This is the separation axiom of a metric. □

Caveat 4.12 (Semantic Shield for Bridge Lemma). The Bridge Lemma is a **metric statement**, not an ontological one. It does not claim that "identity is coupled to experience." It claims that the map Φ satisfies a separation property. Misinterpretation of this lemma as a consciousness claim is outside the mathematical scope of the framework.

5 Main Results

5.1 The Conditional Null-Separation Theorem

Definition 5.1 (Admissible extension witness)**.** For the declared comparison readout $\Phi : Z \to \mathcal{E}$, an extension witness is (S', z_0, z_1, ϵ) with both endpoints in Z, both readout values defined, and $0 < d_Z(z_0, z_1) = \epsilon < \infty$. Its provenance must identify the admitted extension and the embedding/transport that places the endpoints in Z. When $Z = D_{S'}$, z_0 may be a transported point of S, whereas z_1 is a distinct point. Admissibility of this construction and the lower bound on this pair are separate obligations. No finite-cost internal nonzero deformation is inferred from $M_\Omega = +\infty$.

Theorem 5.2 (Conditional Null-Separation Under an Admissible Extension Witness)**.** *Let $\Phi : Z \to \mathcal{E}$ be defined on a declared separated witness pair (z_0, z_1) with $d_Z(z_0, z_1) = \epsilon > 0$ and $d_\mathcal{E}(\Phi(z_0), \Phi(z_1)) \geq C\epsilon$, $C > 0$. Then*

$$\neg(\Phi(z_0) = \varnothing_\phi \ \wedge \ \Phi(z_1) = \varnothing_\phi), \qquad \max_{i=0,1} d_\mathcal{E}(\Phi(z_i), \varnothing_\phi) \geq C\epsilon/2.$$

This applies to a CCR channel if it supplies these data; CCR is unnecessary for the metric conclusion. It excludes joint exact null assignment, not a single stable null endpoint.

Proof. If both images were null their mutual distance would be zero, contradicting the positive lower bound. The triangle inequality bounds their mutual distance by the sum of their distances to null, at most twice their maximum. Rearranging gives the quantitative assertion. The factor $1/2$ is sharp in general: in the coordinatewise ordered positive quadrant with the ℓ^1 metric, values $(a, 0)$ and $(0, a)$ have separation $2a$ and both have null distance a. □

Remark 5.3 (Applicability scope). Theorem 5.2 assumes $\Phi(z_0), \Phi(z_1) \in \mathcal{E}$, not $\perp$. If Φ is undefined on either endpoint, the separated-witness hypothesis of Definition 5.1 is not satisfied and the theorem does not apply.

Caveat 5.4 (v24 correction of the former Step 2). The theorem above does *not* infer a finite-energy non-zero deformation from $M_\Omega = +\infty$. Under the infimum convention for ontological mass, $M_\Omega = +\infty$ means precisely that no finite-energy admissible perturbation produces non-zero deformation. The older Step 2 proof direction is therefore withdrawn. The surviving result is conditional on an independently specified extension witness.

Example 5.5 (External Null Witness in a Common Readout Domain)**.** Fix $k \geq 0$ and $0 < a < 1/2$. Use the weighted circle-prefix history metric with weights 3^{-t}, the undeformed tower I_0, the single-jump tower I_1 (bonding shift $-a$ at level k), and $Z = I_0 \cup I_1$. Set $\beta = 3^{-(k+1)}$. The common readout F takes the initial angle, the angle at $k+1$, and the ternary code of the bits, with codomain metric $\max(\rho_0, \beta\rho_1, |\Delta\text{code}|)$. The two angular blocks give the first two terms of d_w, while the ternary leading term minus its tail bounds the binary part from below. Thus $C = 1$ holds on the entire compact union. Order the codomain by equality together with $e_0 = (0, 0, 0)$ below every element.

Declare internal interventions to be simultaneous rotations of all stages of I_0, with cost the absolute shortest angular shift. Every such map has its inverse and sends I_0 onto itself. This gives infinite resistance in this sector by the Hausdorff convention, despite finite-cost nonidentity transports. The pointwise readout changes under a nonzero rotation. A sector admitting the single-jump deformation instead has a finite resistance witness with projection-defect cost $3^{-k}a$. The two sectors specify different admissible operations.

Declare as a separate extension the addition of the single-jump branch to I_0. It is the union of two compact projective limits; it is also realized by the disjoint union of the two stage towers with branch-preserving projections. Embed its histories in Z by forgetting the tag, which is

injective at the limit since the phases across k distinguish the two branches. The inclusion $j : I_0 \hookrightarrow Z$ preserves the restricted readout exactly. Take $z_0 = j(x)$ with zero phase and all-zero bits and z_1 on the added branch with the same initial phase and bits. Then

$$d_Z(z_0, z_1) = \beta a, \qquad F(z_0) = e_0, \qquad d_{\mathcal{E}}(F(z_1), e_0) = \beta a.$$

The extension is positive in Hausdorff distance, $\beta a/(1+\beta)$, using the two angular blocks. The transported point and the added endpoint are distinct. The scalar ternary readout alone, which discards both angles, assigns them the same value and fails the positive lower bound on this pair. The example realizes a typed extension witness and distinguishes setwise resistance from readout invariance.

5.2 Conditional Consequences

Corollary 5.6 (Conditional Forced Non-Nullity)**.** *If a CCR channel admits an admissible extension witness for Φ and if $\Phi(z_1) = \varnothing_\phi$ for the witness object, then $\Phi(z_0) \neq \varnothing_\phi$. Symmetrically, if $\Phi(z_0) = \varnothing_\phi$, then $\Phi(z_1) \neq \varnothing_\phi$.*

Proof. This is the contrapositive form of Theorem 5.2. □

Remark 5.7 (Ontological Status). This is a **negative-structural and conditional** result. It does not assert that experience exists, that CCR systems are conscious, or that every CCR channel possesses the required witness. It asserts only that, once a separated extension witness is independently supplied, a compatibility operator satisfying (C3) cannot collapse both sides of that witness to the null regime.

6 Stratification of Positive Regimes

6.1 The Positive Regime Space

Definition 6.1 (Positive Regime Space)**.**

$$\mathcal{E}^+ := \mathcal{E} \setminus \{\varnothing_\phi\}$$

with the inherited partial order $\preceq$.

Lemma 6.2 (Witness-Relative No-Minimum Collapse)**.** *The finite positive witness image $A = \{\Phi(z_0), \Phi(z_1)\} \setminus \{\varnothing_\phi\}$ is non-empty. Its infimum, if one exists in the ambient poset, need not be positive or belong to A.*

Proof. Non-emptiness follows from Theorem 5.2. In the complete finite lattice $\{0, a, b, 1\}$ with incomparable a, b and discrete metric, the injective image $\{a, b\}$ is separated, but its infimum is 0. Thus a positive infimum cannot be inferred. □

6.2 Minimal Positive Regime

Theorem 6.3 (Witness-Relative Minimal Positive Regime)**.** *The finite non-empty positive witness image A has at least one minimal element in its inherited partial order. If its two possible elements are comparable, it has a unique least element. These statements require no ambient infimum or order completeness; they concern A, not all of $\mathcal{E}^+$.*

Proof. Starting from any member of a finite non-empty poset, a strictly descending chain must terminate at a minimal member. For a set of at most two comparable members the smaller one is least and unique. The diamond example above has two incomparable minimal members, so uniqueness cannot be inferred without an added condition. □

6.3 Structural Intensity

Definition 6.4 (Structural Intensity)**.** For positive e, distinguish the null distance $r(e) = d_{\mathcal{E}}(e, \varnothing_\phi)$ from the global lower gap

$$\iota(e) = \inf_{f \in \mathcal{E},\ f \prec e} d_{\mathcal{E}}(e, f).$$

For an explicitly audited set A_0 containing e and $\varnothing_\phi$, define the relative gap $\iota_{A_0}(e)$ by the same infimum restricted to $f \in A_0$. All gaps have the units of $d_{\mathcal{E}}$; no conversion to experience or energy is implied.

Corollary 6.5 (Witness-Relative Positive Intensity)**.** *For a minimal positive member e of the finite witness image A and $A_0 = A \cup \{\varnothing_\phi\}$, $\iota_{A_0}(e) = r(e) > 0$. The global $\iota(e)$ need not be positive.*

Proof. The only strictly smaller member of A_0 is null, giving the equality. For $\mathcal{E} = [0, 1]$ with its usual order and metric, $e = 1$ has global gap zero, although the audited set $A_0 = \{0, 1\}$ has relative gap one. □

7 Quantitative Bounds

7.1 Witness Lower Bound

Theorem 7.1 (Witness Intensity Bound)**.** *Under the hypotheses of Theorem 5.2, if exactly one of $\Phi(z_0)$ and $\Phi(z_1)$ is null, then the non-null witness image e satisfies*

$$d_{\mathcal{E}}(e, \varnothing_\phi) \geq C\epsilon > 0.$$

Proof. Apply (C3) to the pair (z_0, z_1). If one side maps to $\varnothing_\phi$ and the other side maps to e, then

$$d_{\mathcal{E}}(e, \varnothing_\phi) = d_{\mathcal{E}}(\Phi(z_0), \Phi(z_1)) \geq C\, d_Z(z_0, z_1) = C\epsilon.$$

□

7.2 Explicit Constants for Canonical Families

Proposition 7.2 (Conditional Profinite Bound)**.** *For a profinite family with a verified extension witness of separation ϵ, one endpoint fixed at $\varnothing_\phi$, and an independently certified witness-level lower constant $C_{\mathrm{prof}} > 0$, the witness image separation satisfies*

$$d_{\mathcal{E}}(e, \varnothing_\phi) \geq C_{\mathrm{prof}}\epsilon.$$

Proof. This is Theorem 7.1 applied with the certified constant C_{prof}. The proposition does not claim that every profinite observable automatically supplies such a constant. □

Proposition 7.3 (Conditional Symbolic Bound)**.** *For a symbolic family with a verified extension witness of separation ϵ, one endpoint fixed at $\varnothing_\phi$, and an independently certified witness-level lower constant $C_{\mathrm{sym}} > 0$, the witness image separation satisfies*

$$d_{\mathcal{E}}(e, \varnothing_\phi) \geq C_{\mathrm{sym}}\epsilon.$$

Proof. This is Theorem 7.1 applied with the certified constant C_{sym}. The proposition does not claim that every symbolic observable automatically supplies such a constant. □

Example 7.4 (p-Adic witness calculation)**.** If a 2-adic model supplies an independently verified witness with $\epsilon = 2^{-1}$ and $C = 1$, with one endpoint mapped to null, then the conditional bound gives $d_{\mathcal{E}}(e, \varnothing_\phi) \geq 1/2$. This is a witness calculation, not a universal theorem that every CCR channel is non-null.

Caveat 7.5 (No lower bound from summability alone). The existence of summable coordinate weights, continuity of a readout, or compactness of a canonical family does not by itself imply a positive lower Lipschitz constant. Cancellation, quotient collapse, non-injective readouts, or an observable that ignores the separated coordinates can make the effective lower constant zero. Every positive value of C_{prof} or C_{sym} used in this paper is therefore a certificate to be supplied for the stated witness pair or witness class, not a free consequence of the ambient family.

8 Witness Audit Layer

The instability theorem is useful only if its hypotheses can be audited without smuggling the desired conclusion into the witness. This section records the minimal audit layer required before the theorem may be used in a concrete argument.

Definition 8.1 (Witness certificate)**.** A witness certificate for a pair (z_0, z_1) consists of the tuple

$$\mathfrak{W} := (z_0, z_1, \epsilon, C, \Phi, \mathcal{E}, \varnothing_\phi, \mathcal{J}),$$

where $\epsilon = d_Z(z_0, z_1) > 0$, $C > 0$ is the certified lower constant for Φ on the audited pair or audited witness class, $\mathcal{J}$ is the provenance record establishing that the endpoint pair comes from a declared typed comparison, rather than an internal relabeling, and all objects are fixed before the null/non-null status is adjudicated.

Definition 8.2 (Witness independence)**.** A witness certificate is conclusion-independent when the construction of the endpoint z_1, the choice of Φ, the metric $d_{\mathcal{E}}$, the null element $\varnothing_\phi$, and the lower constant C are fixed before the endpoint status is adjudicated. Dependence on the upstream CCR structure is allowed; dependence on a preselected endpoint verdict is not.

Proposition 8.3 (Certificate sufficiency)**.** *If a conclusion-independent witness certificate exists for (z_0, z_1), and both endpoints are assigned to $\varnothing_\phi$, then the certificate is inconsistent.*

Proof. The certificate includes the separation condition $d_Z(z_0, z_1) = \epsilon > 0$ and the lower bound

$$d_{\mathcal{E}}(\Phi(z_0), \Phi(z_1)) \geq C\epsilon > 0.$$

If both endpoints are assigned to $\varnothing_\phi$, then the left-hand side is $d_{\mathcal{E}}(\varnothing_\phi, \varnothing_\phi) = 0$, contradicting $C\epsilon > 0$. □

Non-Theorem 8.4 (What a certificate does not prove). A witness certificate does not prove that $\mathcal{I}$ is conscious, living, subjective, morally relevant, biological, human-equivalent, or externally validated. It proves only that the stated pair cannot be jointly collapsed to the same null element under the certified compatibility map.

8.1 Failure Modes of Witness Certification

Failure mode	Formal symptom	Consequence
Absent witness	No typed endpoint z_1 with $d_Z(z_0, z_1) > 0$	Theorem 5.2 cannot be invoked
Zero lower constant	$C = 0$ or no certified positive lower bound	The null assignment is not contradicted by the metric data
Conclusion-dependent witness	z_1 or Φ is chosen after fixing the endpoint verdict	The certificate no longer supplies an independent exclusion of joint nullity
Quotient collapse	$\Phi(z_0) = \Phi(z_1)$ despite $d_Z(z_0, z_1) > 0$	The map may be continuous, but it is not separating on the audited pair
Null redefinition	$\varnothing_\phi$ changes between construction and adjudication	The theorem has no stable target
Estimator-only promotion	Runtime estimates are substituted for Φ without an error margin	The result becomes an implementation claim rather than a theorem

8.2 Robust Decision Margin

Proposition 8.5 (Noise-robust null exclusion)**.** *Let $e_i = \Phi(z_i)$ and estimates $\widehat{e}_i$ satisfy $d_{\mathcal{E}}(e_i, \widehat{e}_i) \leq \eta_i$, $i = 0, 1$. Let $m = C\epsilon$ be a certified true separation. Then*

$$d_{\mathcal{E}}(\widehat{e}_0, \widehat{e}_1) \geq m - \eta_0 - \eta_1.$$

Positive estimated separation follows from $m > \eta_0 + \eta_1$. If null acceptance instead means $d_{\mathcal{E}}(\widehat{e}_i, \varnothing_\phi) \leq r_i$, both estimates cannot be accepted as null when

$$m > \eta_0 + \eta_1 + r_0 + r_1.$$

For equal estimation and acceptance radii η, this sufficient condition is $m > 4\eta$, not merely $m > 2\eta$.

Proof. Apply the triangle inequality through the two estimates. If both estimates lie in their null balls, their mutual distance is at most $r_0 + r_1$, and the true separation is at most $\eta_0 + \eta_1 + r_0 + r_1$. This contradicts the strict condition. As a negative control, true values $(2, 0), (0, 2)$ in $\mathbb{R}^2_+$ and estimates $(1, 0), (0, 1)$ have errors one and true separation $2\sqrt{2} > 2$, yet both estimates lie in the unit null ball. Estimated separation is not the same decision as null-ball exclusion. □

Caveat 8.6 (Implementation boundary). Proposition 8.5 is an error-budget rule for an already specified estimator. It does not assert that the current runtime implements Φ, computes C, or supplies η. Those are separate implementation burdens.

8.3 Negative Controls

1. **Constant-null control.** Set $\Phi(x) = \varnothing_\phi$ for all x. This must fail the lower-bound condition whenever $\epsilon > 0$.

2. **Coordinate-blind control.** Choose a readout that ignores every coordinate on which z_0 and z_1 differ. This tests whether separation is actually carried by the map.

3. **Witness-shuffle control.** Hold Φ fixed and replace z_1 by a typed point not separated from z_0. The theorem must become inapplicable.

4. **Null-relabel control.** Change the declared null element after observing the images. This must be rejected as target drift.

5. **Estimator-margin control.** Vary estimation radii and null-acceptance radii separately. At $m \leq \eta_0+\eta_1$ positive estimated separation is no longer guaranteed; at $m \leq \eta_0+\eta_1+r_0+r_1$ joint null-ball exclusion is no longer guaranteed. The corresponding sufficient margin must fail rather than silently promoting a non-null conclusion.

9 Witness-Relative Closure Theorem

Theorem 9.1 (Conditional Phenomenological Closure)**.** *A declared separated, detected pair has a non-empty positive image and the unanchored lower bound* $\max_i r(\Phi(z_i)) \geq C\epsilon/2$*. Its finite positive image has a minimal member; uniqueness requires comparability. If one endpoint is null, the other is the unique positive witness image, with relative gap and null distance at least* $C\epsilon$*. Temporal stability, global minimal positivity, and physical CCR realizability are not consequences.*

Proof. Combine Theorem 5.2, Theorem 6.3, Corollary 6.5 and the anchored bound of Theorem 7.1. Each conclusion retains its stated hypotheses. □

9.1 Classification Table

System Type	Ontological Mass	Null Regime Permitted?
Degenerate	$M_\Omega = 0$	Not decided by mass alone
Finite-mass	$0 < M_\Omega < \infty$	Not decided by mass alone
CCR without extension witness	$M_\Omega = +\infty$	Not decided here
CCR with separated extension witness	$M_\Omega = +\infty$	Joint null collapse forbidden by Theorem 5.2

10 Null-Fiber Geometry

The null element $\varnothing_\phi$ determines a fiber

$$\mathcal{N}_\Phi := \Phi^{-1}(\{\varnothing_\phi\}) \subseteq Z.$$

The instability theorem can be read as a geometric obstruction on this fiber: a certified witness pair cannot be contained entirely inside $\mathcal{N}_\Phi$.

Definition 10.1 (Null fiber)**.** For a compatibility operator $\Phi : Z \to \mathcal{E}$, the null fiber is the preimage $\mathcal{N}_\Phi = \{x \in Z : \Phi(x) = \varnothing_\phi\}$.

Definition 10.2 (Witness-detecting map)**.** A compatibility operator Φ is witness-detecting on a set $W \subseteq Z \times Z$ when there exists $C_W > 0$ such that

$$d_\mathcal{E}(\Phi(x), \Phi(y)) \geq C_W \, d_Z(x, y)$$

for every $(x, y) \in W$.

Proposition 10.3 (Null-fiber exclusion on detected pairs)**.** *Let W be a witness class on which Φ is witness-detecting with constant $C_W > 0$. If $(x, y) \in W$ and $d_Z(x, y) > 0$, then (x, y) cannot both lie in $\mathcal{N}_\Phi$.*

Proof. If $x, y \in \mathcal{N}_\Phi$, then $\Phi(x) = \Phi(y) = \varnothing_\phi$, so $d_\mathcal{E}(\Phi(x), \Phi(y)) = 0$. Witness detection gives $d_\mathcal{E}(\Phi(x), \Phi(y)) \geq C_W \, d_Z(x, y) > 0$, a contradiction. □

Corollary 10.4 (Diameter obstruction)**.** *If a subset $A \subseteq Z$ contains a detected witness pair (x, y) with $d_Z(x, y) > 0$, then A cannot be contained in $\mathcal{N}_\Phi$.*

Proof. Containment $A \subseteq \mathcal{N}_\Phi$ would place both endpoints of the detected pair in the null fiber, contradicting the previous proposition. □

10.1 Quotient-Collapse Diagnostic

Let $q : Z \to Q$ be a quotient map and let $\Phi = h \circ q$ for some $h : Q \to \mathcal{E}$. If $q(x) = q(y)$ for a separated pair, then $\Phi(x) = \Phi(y)$ for every such h. Therefore no lower-bound certificate can hold on that pair.

Proposition 10.5 (Quotient obstruction). *Suppose $\Phi = h \circ q$ and $q(x) = q(y)$ while $d_Z(x, y) > 0$. Then no constant $C > 0$ can satisfy*

$$d_{\mathcal{E}}(\Phi(x), \Phi(y)) \geq C\, d_Z(x, y).$$

Proof. Since $q(x) = q(y)$, $\Phi(x) = h(q(x)) = h(q(y)) = \Phi(y)$. Hence the left-hand side is zero, while $C\, d_Z(x, y) > 0$ for any $C > 0$. □

10.2 Certificate Stability

Proposition 10.6 (Minimum over a finite witness ledger). *Let $m \geq 1$ and let $W_0 = \{(x_i, y_i)\}_{i=1}^m$ be a finite ledger of separated witness pairs. Suppose each pair has a certified lower constant $C_i > 0$:*

$$d_{\mathcal{E}}(\Phi(x_i), \Phi(y_i)) \geq C_i\, d_Z(x_i, y_i).$$

Then $C_* = \min_i C_i > 0$ is a valid lower constant for the finite ledger.

Proof. For each i, $C_i \geq C_*$. Hence

$$d_{\mathcal{E}}(\Phi(x_i), \Phi(y_i)) \geq C_i\, d_Z(x_i, y_i) \geq C_*\, d_Z(x_i, y_i).$$

Since the ledger is finite and each C_i is positive, $C_* > 0$. □

Proposition 10.7 (Strict margin persistence). *Let $\widehat{m} > 0$ be a certified lower bound on the current estimated pair distance. If a further perturbation changes that distance by at most $\rho < \widehat{m}$, the estimates remain separated. From true margin $m = C\epsilon$ and estimator radii, one may use $\widehat{m} = m - \eta_0 - \eta_1$ only when it is positive.*

Proof. The new estimated distance is at least $\widehat{m} - \rho > 0$. An unadjusted true margin cannot be substituted for the current estimated one. □

10.3 Operational Checklist

For a claimed null-fiber exclusion, the certificate record must identify:

1. the null element $\varnothing_\phi$;
2. the endpoints (z_0, z_1) or the audited witness class W;
3. the separation value ϵ or the family of separations;
4. the map Φ and the metric $d_{\mathcal{E}}$;
5. the lower constant C or the finite-ledger constant C_*;
6. the estimator margin, when an estimator is used;
7. the quotient maps that have been ruled out as collapse explanations.

11 Comparison with Existing Frameworks

11.1 Integrated Information Theory (IIT)

	IIT	**This Framework**
Claims consciousness exists	Yes	No
Defines Φ	Information integration	Structural map
Quantitative measure	Φ (bits)	$\iota(e_\star)$ (metric)
Metaphysical commitment	Panpsychism	None
Falsifiable predictions	Debated	None (structural)

11.2 Global Workspace Theory (GWT)

Our framework is orthogonal to GWT. We make no claims about attention, broadcast, or access consciousness. The structural results concern identity stability, not cognitive architecture.

11.3 Higher-Order Theories

We do not model representations, meta-cognition, or self-awareness. The space $\mathcal{E}$ contains regimes, not mental states.

11.4 Mathematical Foundations (Chalmers, Nagel)

Our framework provides formal tools that *could* be used in philosophical analysis, but we do not claim to resolve the "hard problem." The conditional exclusion of joint exact nullity is a metric fact, not an explanation of subjective experience.

12 Limitations and Open Problems

12.1 Limitations

1. **No claim about qualia types, contents, or subjectivity.**
2. **No claim that all non-CCR systems are null.**
3. **Result is structural necessity, not phenomenological doctrine.**
4. **All results are conditional on the axioms of Papers I–II.**
5. **The physical existence of CCR systems is unknown.**

12.2 Open Problems

1. **Characterization of $\mathcal{E}^+$:** What is the full structure of positive regimes?
2. **Regime dynamics:** How do regimes evolve under channel evolution?
3. **Physical CCR:** Do CCR systems exist in physics (quantum gravity, etc.)?
4. **Biological CCR:** Could biological systems approximate CCR?
5. **Computational bounds:** Sharper non-simulability results?

12.3 Foundation-Level Gaps and Non-Claims

The following gaps remain open in the current corpus and are documented here to prevent over-reading of the separation theorem.

Non-Theorem 12.1 (Topology consistency gap). Papers I–II assume both compact Hausdorff and Polish structures on state spaces, but do not prove their consistency for every QICN instantiation. The inverse limit $\mathcal{I}$ uses compactness and closed-subspace arguments, while the metric d_w uses a metric regularity structure. Every compact metric space is Polish, so the composition is consistent when each S_t is compact metric. Non-compact Polish spaces such as $\mathbb{R}^n$ require additional hypotheses and are not automatically covered.

Non-Theorem 12.2 (Channel–projection composition gap). The observable channel $\mathcal{C}$ and the projective projections $\pi_{t+1\to t} : S_{t+1} \to S_t$ are not yet connected by a fully explicit composition law. Any detectability theorem that moves from observed responses to inverse-limit deformation requires such a composition. At present, that composition is an assumption to be specified for each concrete protocol, not a derived universal theorem.

Non-Theorem 12.3 (Null-regime exclusion requires external witness). The instability theorem (Theorem 5.2) requires a separated external witness (z_0, z_1, ϵ, C). CCR alone does not produce such a witness. The theorem is witness-relative: if a CCR system has a typed external witness with lower Lipschitz constant $C > 0$, then both endpoints cannot simultaneously be null. It does not state that CCR by itself implies non-nullity.

Non-Theorem 12.4 (Estimator verification gap). No Lipschitz constant K_i, fiber oscillation $\omega_i(y)$, estimation error bound ε_i, or decision margin Δ^* has been computed for any QICN invariant. The projection-invariant bridge theorem is conditional on these quantities being verified, and all four material bridge hypotheses remain unverified.

13 Conclusion

We have proven that CCR architectures equipped with a separated external witness and the lower-bound regularity hypotheses structurally exclude joint null collapse. In the present model, the hypothesis "CCR plus typed witness with joint null assignment" is formally incoherent.

This is a structural exclusion result. It does not define phenomenology or turn the theorem into an empirical consciousness attribution.

13.1 Summary of Contributions

1. **Phenomenological Instability Theorem:** joint null collapse is excluded under the typed witness hypotheses.
2. **Minimal Positive Regime:** witness-relative positive regimes can be characterized on the finite audited image; uniqueness requires comparability.
3. **Quantitative Bounds:** Null distance is at least $C\epsilon$ for a null-anchored witness, and relative gap agrees with that distance at a minimal finite witness image.
4. **Canonical Examples:** Profinite and symbolic witness bounds are conditional; summability does not certify infinite mass.
5. **Scope Conditions:** Clear scope delimitation throughout.

A Profinite Canonical Family

A.1 Projective System

Let $\{G_n, \pi_{n+1\to n}\}_{n\in\mathbb{N}}$ be a projective tower of finite groups with surjective morphisms. Define:

$$G := \varprojlim G_n = \left\{ (x_n) \in \prod_n G_n : \pi_{n+1\to n}(x_{n+1}) = x_n \right\}$$

Example A.1 (p-Adic Integers)**.** $G_n = \mathbb{Z}/p^n\mathbb{Z}$, $\pi_{n+1\to n}$ is reduction mod p^n. Then $G = \mathbb{Z}_p$.

A.2 Profinite Metric

Define the profinite metric:

$$d_G(x, y) := \sum_{n\geq 0} \alpha_n \mathbf{1}_{x_n \neq y_n}$$

where $\alpha_n = 2^{-(n+1)}$ (standard choice).

For a projective group tower, coordinate disagreement persists at all higher levels, so this weighted-tail distance is an ultrametric. Its tails vanish uniformly; (G, d_G) is compact, complete, and totally disconnected.

A.3 CCR Verification

The following cost control uses state comparisons in the declared canonical domain. It does not infer a whole-object deformation metric or a physical energy law.

Proposition A.2. *Summable coordinate-change costs do not certify CCR. If an admitted transformation changes infinitely many coordinates and costs $\sum_{n\in J} w_n$ with $\sum_n w_n < \infty$, its cost is finite.*

Proof. The cost of any subset is bounded by the total convergent sum. For example, addition of one sends the zero element of $\mathbb{Z}_2$ to a distinct compatible element and changes every residue coordinate at positive levels; this has finite coordinate-change cost under the declared summable rule. If this intervention is admitted with positive deformation, the infimum mass is finite. Infinite mass needs a different justified intervention/cost restriction, not just profinite topology. □

A.4 Phenomenological Operator

Define $\Phi_f : G \to \mathcal{E}$ by:

$$\Phi_f(x) := f\left(\sum_{n\geq 0} \beta_n \varphi_n(x_n) \right)$$

where $\varphi_n : G_n \to [0, 1]$ are 1-Lipschitz (w.r.t. discrete metric), $\beta_n \geq 0$, $\sum_n \beta_n < \infty$, and $f : \mathbb{R} \to \mathcal{E}$ is L_f-Lipschitz.

A.5 Coupling Bound

Theorem A.3 (Certified Profinite Witness Bound)**.** *Assume that the chosen profinite readout has been certified on the audited witness class by a positive lower constant $C_{\text{prof}} > 0$:*

$$d_{\mathcal{E}}(\Phi_f(x), \Phi_f(\tilde{x})) \geq C_{\text{prof}} \cdot d_G(x, \tilde{x})$$

for the witness pair or witness neighborhood under review. Then any witness pair with $d_G(x, \tilde{x}) = \epsilon > 0$ satisfies

$$d_{\mathcal{E}}(\Phi_f(x), \Phi_f(\tilde{x})) \geq C_{\text{prof}}\epsilon.$$

Proof. This is the certified lower-bound hypothesis applied to the witness pair. □

Remark A.4 (Why certification is necessary). The assumptions that the coordinate functions are Lipschitz and that the coefficients are summable give continuity, not a lower bound. A lower bound requires additional structure such as injectivity on the witness coordinates, no cancellation on the audited pair, a positive margin over the relevant finite support, or an independent estimator certificate.

A.6 Intensity Bound

For a certified witness separation $\epsilon > 0$ and one null endpoint, the other image satisfies

$$r(e) \geq C_{\text{prof}}\epsilon.$$

B Symbolic Canonical Family

B.1 Subshift of Finite Type

Let $\mathcal{A} = \{1, \ldots, k\}$ be a finite alphabet and A a $k \times k$ matrix with entries in $\{0, 1\}$. The **subshift of finite type (SFT)** is:

$$\Sigma_A := \{x \in \mathcal{A}^{\mathbb{Z}} : A_{x_j, x_{j+1}} = 1 \text{ for all } j \in \mathbb{Z}\}$$

Example B.1 (Full Shift)**.** $A_{ij} = 1$ for all i, j. Then $\Sigma_A = \{0, 1\}^{\mathbb{Z}}$.

B.2 Bowen Metric

$$d_B(x, y) := \sum_{j \in \mathbb{Z}} \gamma_{|j|} \mathbf{1}_{x_j \neq y_j}$$

where $\gamma_m = 2^{-(m+1)}$ (standard choice).

The space (Σ_A, d_B) is compact and complete.

B.3 CCR Verification

The following cost control uses state comparisons in the declared canonical domain. It does not infer a whole-object deformation metric or a physical energy law.

Proposition B.2. *Symbolic compactness and summable coordinate weights do not certify CCR under coordinate-change cost.*

Proof. In the binary full shift the coordinatewise complement is admitted if the intervention class includes it. It changes infinitely many symbols, at total cost at most $\sum_{j \in \mathbb{Z}} w_j < \infty$, and maps one compatible sequence to another distinct one. This is a finite-cost positive deformation, so the infinite-mass conclusion fails for that intervention model. A restricted SFT requires its own admitted transformation; no universal CCR certificate follows. □

B.4 Phenomenological Operator

Define $\Phi_g : \Sigma_A \to \mathcal{E}$ by:

$$\Phi_g(x) := g\left(\sum_{j \in \mathbb{Z}} \delta_j \psi_j(x)\right)$$

where $\psi_j : \Sigma_A \to [0, 1]$ are cylinder functions (depending on x_j only), $\delta_j \geq 0$, $\sum_j \delta_j < \infty$, and $g : \mathbb{R} \to \mathcal{E}$ is L_g-Lipschitz.

B.5 Coupling Bound

Theorem B.3 (Certified Symbolic Witness Bound)**.** *Assume that the chosen symbolic readout has been certified on the audited witness class by a positive lower constant $C_{\mathrm{sym}} > 0$:*

$$d_{\mathcal{E}}(\Phi_g(x), \Phi_g(\tilde{x})) \geq C_{\mathrm{sym}} \cdot d_B(x, \tilde{x})$$

for the witness pair or witness neighborhood under review. Then any witness pair with $d_B(x, \tilde{x}) = \epsilon > 0$ satisfies

$$d_{\mathcal{E}}(\Phi_g(x), \Phi_g(\tilde{x})) \geq C_{\mathrm{sym}}\epsilon.$$

Proof. This is the certified lower-bound hypothesis applied to the witness pair. □

C Hereditary Rigidity

Theorem C.1 (Rigidity Inheritance)**.** *Suppose a readout $\Phi : Z \to \mathcal{E}$ has lower constant C on a declared pair class. Let $a : Z' \to Z$ and $b : \mathcal{E} \to \mathcal{E}'$ satisfy lower bounds with constants $c_a, c_b > 0$, and set $\Phi' = b \circ \Phi \circ a$. Then Φ' has lower constant $c_a c_b C$ on the pulled-back class. To inherit CCR additionally require that every finite-cost positive deformation in the new sector pulls back to a finite-cost positive deformation in the original sector. Summability alone supplies neither assumption.*

Proof. Successive application of the three lower bounds gives $d_{\mathcal{E}'}(\Phi'(x), \Phi'(y)) \geq c_b c_a C d_{Z'}(x, y)$. If the original sector admits no finite-cost positive deformation, the specified pullback property rules out one in the new sector by contradiction. Stability moduli and null anchors, if required, must also be transported explicitly. □

D Universal Factorization

Theorem D.1 (Categorical Universality)**.** *If $\Phi : Z \to \mathcal{E}$ has a positive global lower bound, it is injective. For any $\Psi : Z \to F$ there is a unique factor map $u : \Phi(Z) \to F$ with $u \circ \Phi = \Psi$. If Ψ is L-Lipschitz, u is L/C-Lipschitz. No unique extension to all of $\mathcal{E}$ follows.*

Proof. Define $u(\Phi(z)) = \Psi(z)$. Injectivity gives well-definedness and uniqueness on the image. The inverse of Φ on its image is $1/C$-Lipschitz, which gives the stated bound. At a point of $\mathcal{E} \setminus \Phi(Z)$ the commuting identity specifies no value, and a continuous extension may require additional conditions. □

Consequence: All representations of the same declared domain factor setwise through a faithful image. A categorical claim additionally fixes objects and morphisms: with continuous maps and continuous faithful Φ having continuous inverse on its image, that image representation is initial. This does not classify arbitrary ambient codomains or physical systems.

E Non-Simulability Bounds

E.1 Simulation Error

Fix the target readout and the information available to the decoder. On the binary prefix tower, let p_N reveal bits $0, \ldots, N$ and let $F(b) = \sum_{j \geq 0} 2b_j/3^{j+1}$. Define

$$\mathrm{err}_N = \inf_{g:\{0,1\}^{N+1} \to \mathbb{R}} \sup_b |F(b) - g(p_N b)|.$$

The target is not optimized away, and the resource restriction is a canonical prefix, not finite real dimension.

Theorem E.1 (Simulation Lower Bound)**.** *For the fixed target and prefix class above,*

$$\mathrm{err}_N = \frac{1}{2\,3^{N+1}} > 0, \qquad \lim_{N\to\infty} \mathrm{err}_N = 0.$$

Thus exact reconstruction fails at every fixed prefix budget, while arbitrarily accurate approximation is possible as the budget grows.

Proof. Every prefix fiber has two completions with tails all zero or all one, whose images differ by $3^{-(N+1)}$. A single decoder value incurs error at least half that distance on one extreme. The midpoint of the two extremes attains this bound for the entire fiber. This proves equality. The bound tends to zero; it cannot yield a positive limiting error or a prohibition on finite-dimensional real encodings. □

Corollary E.2. *In this prefix model, worst-case error at most ε requires $1/(2\,3^{N+1}) \le \varepsilon$. Hence N must diverge as $\varepsilon \downarrow 0$. This does not forbid a real-valued encoding with unbounded precision.*

F Final Structural Closure

Theorem F.1 (Closure by Non-Alternatives)**.** *The following obligations are jointly sufficient for the corresponding structural conclusions: an admitted separated pair with a certified lower constant gives joint null exclusion; a null anchor gives the stronger anchored bound; faithful global readout gives image factorization; declared prefix access gives the approximation bound above; and the explicit pullback conditions of Appendix C give inheritance. A competing model satisfying those same obligations obeys those same conclusions. Neither CCR realizability nor satisfaction of all obligations is inferred from canonical topology alone.*

Proof. Apply the witness theorem, the anchored bound, and the corrected results of Appendices C–E with their hypotheses. They are independent implications; no circular existence assertion is used to supply their premises. □

Remark F.2 (Structural Import). The internal sector of the example meets the mass-only CCR definition of this paper. The open program is to realize the stronger preservation conditions, witness geometry, costs, transport and readout for the intended target class. Example 5.5 realizes the typed witness and the required comparison geometry for a specified sector. Temporal instability of an individual null state requires a separate dynamical result.

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